# Module - 1

Types of ML models.

→ Supervised learning
→ Semi Supervised
→ Unsupervised
→ Reinforcement

## Supervised learning

* learning from past experience
* Use labelled data

| Age | Has_job | Own_house | Credit-Rating | Class |
|---|---|---|---|---|
| Young | false | false | good | ? |

Focus: learn a target fn that can be used to predict the values of discrete class attribute eg. approve or not approve

Data: A set of data records (also called eg, instances or cases) described by:

i) K attributes: $A_1, A_2, \ldots, A_K$.
ii) a class: Each eg is labelled with pre-defined class.

Goal: To Derive a classification model from the data that can be used to predict the classes of new cases/instances.

Unsupervised Learning

* Class labels of data are unknown.

Classification & Regression

2 steps → Learning
→ Testing

Accuracy = $\frac{\text{No. of correct classification}}{\text{Total no. of test cases}}$

Clustering & Associations

Semisupervised - some portion is labelled, but most is unlabelled

Reinforcement learning - learn from feedbacks - no labelled data

Maximum Likelihood Estimation

$$P(A/B) = \frac{P(B/A) * P(A)}{P(B)}$$

→ Calculating the prob. of a hypothesis (H) being true given that a certain event (E) has happened.

ie, $$P(H/E) = \frac{P(E/H) * P(H)}{P(E)}.$$

$$P(E) = P(H)\,P(E/H) + P(\sim H)\,P(E/\sim H)$$

$$P(H/E) = \frac{99 * 0.1}{0.1 \times 0.99 + \cancel{0.99 \times 0.90}\ 90. \times 1}$$

$$= \frac{99 \times 0.1}{99 \times 0.1 + 99.9 \times 1} = 9\%$$

$\rightarrow \theta \rightarrow$ set of parameters

$\therefore L(\theta) = P(X|\theta) = P(x_1|\theta) * P(x_2|\theta) * \cdots P(x_n|\theta)$

$= \prod_{i=1}^{n} P(x_i|\theta)$

$\rightarrow$ applicable only on independent dataset

$$\frac{d}{d\theta}\left[\log L(\theta)\right] = 0$$

Suppose that X is a discrete random variable with the following pmf where $\theta$ $0 \le \theta \le 1$ is a parameter. The following 10 independent observations are made from such a distribution

[3, 0, 2, 1, 3, 2, 1, 0, 2, 1]

What is the max likelihood of $\theta$.

| X | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(X) | $2\theta/3$ | $\theta/3$ | $2(1-\theta)/3$ | $(1-\theta)/3$ |

$$\frac{2\theta}{3} + \theta/3 + 2(1-\theta)/3 + (1-\theta)/3 = 1$$

$\therefore$ it is valid pmf.

$L(\theta) = P(X|\theta) = P(x=3|\theta)^2 \cdot P(x=0|\theta)^2 \cdot P(x=2|\theta)^3 \cdot P(x=1|\theta)^3$.

$\log L(\theta) = 2\log P(x=3|\theta) + 2\log P(x=0|\theta) + 3\log(P(x=2|\theta)) + 3\log(P(x=1|\theta))$

$= 2\log(1-\theta/3) + 2\log(2\theta/3) + 3\log(2(1-\theta)/3) + 3\log(\theta/3)$.

$\frac{d}{dx}\left[\log L(\theta)\right] = \cancel{\frac{2\times 3}{1-\theta} + \frac{2\times 3}{2\theta}}$

$= \cancel{\frac{2(1-\theta)}{3}}$

$= \cancel{\frac{2\times 3}{1-\theta}}$

$= 2\left[\log(1-\theta) - \log 3\right] + \log(2\theta) - \log 3]$.

$+ 3\left[\log(2(1-\theta)) - \log 3 + \log\theta - \log 3\right]$

$= 2\left[\frac{1}{1-\theta} \times -1 + \frac{1}{2\theta} \times 2\right] + 3\left[\frac{1}{2(1-\theta)} \times -2 + \frac{1}{\theta}\right]$

$$= \frac{-2}{1-\theta} + \frac{2}{\theta} - \frac{3}{1-\theta} + \frac{3}{\theta}$$

$$+ \frac{5}{\theta} - \frac{5}{1-\theta} = 0$$

$$\frac{1}{\theta} - \frac{1}{1-\theta} = 0$$

$$1-\theta - \theta = 0$$

$$\theta = 1/2$$

Q $(3, 0, 2, 1, 3)$.

$$L(\theta) = P(X|\theta) = P(x=3|\theta)^2 * P(x=0|\theta) * P(x=2|\theta) * P(x=1|\theta)$$

$$\log[L(\theta)] = 2\log\left(\frac{1-\theta}{3}\right) + \log(2\theta/3) + \log\left(\frac{2(1-\theta)}{3}\right) + \log(\theta/3)$$

$\theta = 0.4$

$$\frac{-3}{1-\theta} + \frac{2}{\theta} = 0$$

$-3\theta + 2 - 2\theta = 0$

$$= 2\left[\log(1-\theta) - \log 3\right] + \log 2\theta - \log 3 + \log(2(1-\theta)) - \log 3 + \log\theta - \log 3$$

$$= \frac{-2}{1-\theta} + \frac{1}{\theta} + \frac{1 \times -2}{2(1-\theta)} + 1/\theta$$

$$\frac{-3}{1-\theta} + \frac{2}{\theta} = 0 \; ; \quad -3\theta + 2 - 2\theta = 0$$

$$\theta = 2/5 = 0.4$$

The baby weights are normally distributed with mean $\mu$ & SD $\sigma$. U have 3 babies with wts 7, 8, 9. What is the MLE for $\mu$.

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$L(\mu) = P(x=7|\mu) * P(x=8|\mu) * P(x=9|\mu)$$

$$= \left(\frac{1}{\sigma\sqrt{2\pi}}\right) e^{-\frac{(7-\mu)^2}{2\sigma^2}} * \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(8-\mu)^2}{2\sigma^2}} * \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(9-\mu)^2}{2\sigma^2}}$$

$$= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^3 e^{\left(-\frac{(7-\mu)^2+(8-\mu)^2+(9-\mu)^2}{2\sigma^2}\right)}$$

$$\cancel{\log(L(\mu)) = -\left(\frac{(7-\mu)^2+(8-\mu)^2+(9-\mu)^2}{2\sigma^2}\right)\log\left(\frac{1}{\sigma\sqrt{2\pi}}\right)^3}$$

$$\cancel{= -\left(\frac{(7-\mu)^2+(8-\mu)^2+(9-\mu)^2}{2\sigma^2}\right)^*}$$

$$\log(L(\mu)) = \log\left(\frac{1}{\sigma\sqrt{2\pi}}\right)^3 + -\frac{((7-\mu)^2+(8-\mu)^2+(9-\mu)^2)}{2\sigma^2}$$

$$= -3\log(\sigma) - 3\log(\sqrt{2\pi}) - \frac{((7-\mu)^2+(8-\mu)^2+(9-\mu)^2)}{2\sigma^2}$$

$$= -3\log(\sigma) - \frac{3}{2}\log(2\pi) - \frac{((7-\mu)^2+(8-\mu)^2+(9-\mu)^2)}{2\sigma^2}$$

$$\frac{d}{d\mu}(L(\mu)) = 0 - 0 - \frac{1}{2\sigma^2}\left(2(7-\mu)*-1 + 2(8-\mu)*-1 + 2(9-\mu)*-1\right) = 0$$

$$= \frac{1}{\sigma^2}(7-\mu+8-\mu+9-\mu) = 0$$

$$\frac{24-3\mu}{\sigma^2} = 0$$

$$\mu = 24/3 = 8$$

Q A coin is flipped 100 times given that there are 55 heads. Find the MLE for given coin

$$P(x) = nC_x\, p^x (1-p)^{n-x}$$

$$n = 100 \qquad x = 55$$

$$P(m) = 100C_{55}\; p^{55} (1-p)^{45}$$

$$\log(P(x)) = \log(100C55) + 55\log p + 45\log(1-p)$$

$$\frac{d}{dx}(\log(P(x))) = \frac{55}{p} + \frac{45}{1-p}\, x-1 = 0$$

$$55(1-p) - 45p = 0$$

$$55 = 100p$$

$$p = \frac{11}{20}$$

Q. Find the likelihood fn for $x_i \sim$ Binomial $(3, \theta)$ and we have observations $x_1, x_2, x_3, x_4 = (1, 3, 2, 2)$

$$P(x) = n_{C_x}\, p^x (1-p)^{n-x}$$

$$L(\theta) = 3C_1\, \theta^1 (1-\theta)^2 \times 3C_3\, \theta^3 (1-\theta)^0 \times (3C_2\, \theta^2 (1-\theta)^1)^2$$

$$\log(L(\theta)) = \log(3C_1) + \log(\theta) + 2\log(1-\theta) + 3\log\theta + 2(\log(3C_2) + 2\log\theta + \log(1-\theta))$$

$$\frac{d}{d\theta}(\log(L(\theta)))$$

$$= \frac{1}{\theta} + \frac{2x-1}{1-\theta} + \frac{3}{\theta} + \frac{4}{\theta} + \frac{2x-1}{1-\theta} = 0$$

$$= \frac{8}{\theta} + \frac{-4}{1-\theta} = 0$$

$$8 - 8\theta - 4\theta = 0$$

$$\theta = \underline{2/3}$$

Q. $x_i \sim$ Exponential $(\theta)$

$(x_1, x_2, x_3, x_4) = (1.23, 3.32, 1.98, 2.12)$

$$f(x_i|\theta) = \theta \cdot e^{-\theta \cdot x}$$

$$L(\theta) = \theta \cdot e^{-1.23\theta} \times \theta e^{-3.32\theta} \times \theta e^{-1.98\theta} \times \theta e^{-2.12\theta}$$

$$= \theta^4 e^{-8.65\theta}$$

$$\log(L(\theta)) = 4\log\theta + -8.65\theta$$

$$\frac{d}{dx}(\log(L\theta)) = \frac{4}{\theta} - 8.65 = 0$$

$$\theta = \frac{4}{8.65} = \underline{0.46}$$

Q. Find the MLE for Bernoulli density fn

$$P(x|p) = p^x (1-p)^{1-x}$$

$$L(p) = p(x_1, x_2 \ldots x_n | p)$$

$$= p(x_1|p) \cdot p(x_2|p) \cdots p(x_n|p)$$

$$= p^{x_1}(1-p)^{(1-x_1)} * p^{x_2}(1-p)^{(1-x_2)} * \dots * p^{x_n}(1-p)^{(1-x_n)}$$

$$\log(L(p)) = x_1\log p - (1-x_1)\log(1-p) + x_2\log p - (1-x_2)\log(1-p) + \dots + x_n - (1-x_n)\log(1-p)$$

$$\frac{d}{dp}(\log(L(p))) = \frac{x_1}{p} - \frac{(1-x_1)}{1-p} + \dots + \frac{x_n}{p} + \frac{(1-x_n)}{1-p}$$

$$= \frac{x_1+x_2+\dots+x_n}{p} - \frac{n-(x_1+x_2+\dots+x_n)}{(1-p)} = 0$$

$$(x_1+x_2+\dots+x_n)(1-p) - [n-(x_1+x_2+\dots+x_n)]p = 0$$

$$(x_1+x_2+\dots+x_n) - \cancel{(x_1+\dots+x_n)p} - np + \cancel{(x_1+\dots+x_n)p} = 0$$

$$p = \frac{x_1+x_2+\dots+x_n}{n}$$

$$p = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Q. Poisson

Q. Geometric distribution

Q. Poisson distribution

$$P(x|\theta) = \frac{e^{-\lambda}\lambda^x}{x!}$$

~~P(X = x)~~

$$L(\lambda) = P(x_1, x_2 \dots x_n|\lambda)$$

$$= P(x_1|\lambda) * P(x_2|\lambda) * \dots * P(x_n|\lambda)$$

$$= \frac{e^{-\lambda}\lambda^{x_1}}{x_1!} * \frac{e^{-\lambda}\lambda^{x_2}}{x_2!} * \dots * \frac{e^{-\lambda}\lambda^{x_n}}{x_n!}$$

$$\log(L(\lambda)) = -\lambda\log e + x_1\log\lambda - \log(x_1!) + \dots + -\lambda\log e + x_n\log\lambda - \log(x_n!)$$

$$= -n\lambda + \log\lambda(x_1+x_2+\dots+x_n) - [\log x_1! + \dots \log x_n!]$$

$$\frac{d}{d\lambda}(\log L(\lambda)) = -n + \frac{x_1+x_2+\dots+x_n}{\lambda} - 0 = 0$$

$$\lambda = \frac{x_1+x_2+\dots+x_n}{n}$$

$$\lambda = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$$

Q. Geometric distribution

$$P(x|p) = (1-p)^{x-1} * p$$

$$L(p) = P(x_1, x_2 \dots x_n)$$
$$= (1-p)^{x_1-1} * p * (1-p)^{x_2-1} p * \dots * (1-p)^{x_n-1}$$

$$L(p) = p^n * (1-p)^{x_1-1} * (1-p)^{x_2-1} * \dots * (1-p)^{x_n-1}$$

$$\log(L(p)) = n\log(p) + (x_1-1)\log(1-p) + \dots + (x_n-1)\log(1-p)$$

$$\log(L(p)) = n\log(p) + \log(1-p)\,[x_1 + \dots + x_n - n]$$

$$\frac{d}{dp}\log(L(p)) = \frac{n}{p} + \frac{x_1+x_2+\dots x_n - n}{1-p} * -1 = 0$$

$$n(1-p) - (x_1+x_2+\dots x_n)p - np) = 0$$

$$n - np - (x_1+x_2+\dots+x_n)*p + np = 0$$

$$\therefore p = \frac{n}{x_1+x_2+\dots+x_n}$$

$$p = \frac{1}{\bar{x}}$$

MAP (Maximum Aposteriori Probability).

$$\underset{\text{posterior prob.}}{P(A/B)} = \frac{\overset{\text{likelihood}}{P(B/A)} * P(A)}{P(B)} \text{ prior info}$$

→ It uses prior data

$$P(\theta/D) = \frac{P(D/\theta) * P(\theta)}{P(D)} \qquad \theta = \text{hypothesis}$$

ie $P(\theta/D) = P(D/\theta) * P(\theta)$.

Q 2) Find the MLE ~~and ma~~ estimator for the mean of a normal distribution assuming that we have n samples $x_1, x_2, \dots, x_n$ independently drawn from a normal distribution with non-variance $\sigma^2$ & unknown mean $\mu$.

ii) Derive the MAP estimate for the mean $\mu$ assuming that the prior distribution of mean is itself a normal dist with mean $\nu$ & variance $\beta^2$.

iii) Observe in which condition MLE & MAP behaves the same.

~~MLE~~

i) $$P(x|\mu) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$L(\mu) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_1-\mu)^2+(x_2-\mu)^2+\dots+(x_n-\mu)^2}{2\sigma^2}}$$

$\log(L(\mu)) =$

$$\log\left(\frac{1}{\sigma\sqrt{2\pi}}\right) + -\left[\frac{(x_1-\mu)^2+(x_2-\mu)^2+\dots+(x_n-\mu)^2}{2\sigma^2}\right]$$

$$\frac{-1}{2\sigma^2}\left[2(x_1-\mu)*-1 + \dots + 2(x_n-\mu)*-1\right] =$$

$$\frac{*1}{2\sigma^2}\, 2\left[(x_1+x_2+\dots+x_n) - n\mu\right] = 0$$

$$\mu = \frac{x_1+x_2+\dots+x_n}{n}$$

(ii) $$P(\theta|D) = P(D|\theta) * P(\theta).$$

$$P(\mu|D) = P(D|\mu) * P(\mu).$$

$$= \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}} * \frac{1}{\beta\sqrt{2\pi}} e^{-\frac{(\mu-\nu)^2}{2\beta^2}}$$

$$= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_1-\mu)^2+\dots+(x_n-\mu)^2}{2\sigma^2}} * \frac{1}{\beta\sqrt{2\pi}} e^{-\frac{(\mu-\nu)}{2\beta^2}}$$

$$= \log\left(\frac{1}{\sigma\sqrt{2\pi}}\right) + -\frac{(x_1-\mu)^2+\dots+(x_n-\mu)^2}{2\sigma^2} + \log\left(\frac{1}{\beta\sqrt{2\pi}}\right) + -\frac{(\mu-\nu)^2}{2\beta^2}$$

$$= 0 - \frac{1}{2\sigma^2} \times 2(x_1+x_2+\dots+x_n - n\mu) + 0 + \frac{-1}{2\beta^2} \times 1 - 0$$

$$\frac{x_1+x_2+\dots+x_n-n\mu}{\sigma^2} - \frac{1}{2\beta^2} = 0$$

$$2\beta^2(x_1+x_2+\dots+x_n - n\mu) - \sigma^2 = 0$$

$$2\beta^2 \sum_{i=1}^{n} x_i - n\mu \times 2\beta^2 - \sigma^2 = 0$$

$$\mu = \frac{2\beta^2 \sum_{i=1}^{n} x_i - \sigma^2}{2\beta^2 n}$$

$$= \bar{x} \cancel{- \frac{\sigma^2}{2\beta^2 n}}$$

$$= \frac{\sigma^2 v + \beta^2 \sum_{i=1}^{n} x_i}{\sigma^2 + n\beta^2}$$

$$= 0 + \frac{-1}{2\sigma^2} \times 2(x_1 + \dots + x_n) - n\mu + 0 + \frac{-2(\mu - v)}{2\beta^2} = 0$$

$$\frac{x_1 + x_2 + \dots + x_n - n\mu}{\sigma^2} - \frac{2(\mu - v)}{2\beta^2} = 0$$

$$(x_1 + x_2 + \dots + x_n)\beta^2 - n\mu\beta^2 - \sigma^2\mu + \sigma^2 v = 0$$

$$\sigma^2 v + \beta^2 \sum_{i=1}^{n} x_i = \mu(n\beta^2 + \sigma^2)$$

$$\therefore \mu = \frac{\sigma^2 v + \beta^2 \sum_{i=1}^{n} x_i}{n\beta^2 + \sigma^2}$$

Q. If the data $x_1, \dots x_n$ are iid Poisson $(\lambda)$, then a gamma $(\alpha, \beta)$ prior on $\lambda$ is a conjugate prior

$$P(\lambda|x) = P(x|\lambda) * P(\lambda)$$

$$\therefore P(x|\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$= \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod_{i=1}^{n} (x_i)!}$$

$$\text{ie } P(x|\lambda) \propto e^{-n\lambda} \lambda^{\sum x_i}$$

Gamma prior: $P(\lambda) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \lambda > 0.$

$$\text{ie } P(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$\therefore P(\lambda|x) \propto e^{-n\lambda} \lambda^{\sum x_i} * \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$\text{ie } P(\lambda|x) \propto \lambda^{\sum x_i + \alpha - 1} e^{-(n+\beta)\lambda}$$

ie Posterior probability is gamma $(\sum x_i + \alpha, n + \beta)$

$\therefore$ it is a conjugate prior since posterior prob. is in the same family

Q. Consider the geometric distribution, which has pdf

$$P(X=k) = (1-\theta)^{k-1} \cdot \theta$$

Assume that n iids are drawn from that distribution

a) Find the MLE for θ?

b) Let θ have a beta prior distribution. What is the posterior distribution of θ.

(i) $P(x|p) = (1-p)^{x-1} * p$

$L(\theta) = p^n * (1-p)^{x_1-1} * (1-p)^{x_2-1} * \dots * (1-p)^{x_n}$

$\frac{d}{dp}(\log(L(p))) = \frac{n}{p} + \frac{x_1+x_2+\dots+x_n-n}{1-p} * -1 = 0$

$n(1-p) + -((x_1+x_2 \dots + x_n)\theta - np) = 0$

$\therefore \theta \; \frac{n}{x_1+x_2+\dots+x_n} = \frac{1}{\bar{x}}$

(ii) $P(\theta|x) = P(x|\theta) * P(\theta)$

$P(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}$

$B(\alpha,\beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1} dt$

$P(\theta|x) = \theta^n * (1-\theta)^{x_1-1} * \dots * (1-\theta)^{x_n-1} * \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}$

$\log(L(\theta))$

$= n\log\theta + (x_1+x_2+\dots+x_n-n)\log(1-\theta) + (\alpha-1)\log\theta + (\beta-1)\log(1-\theta) - \log(B(\alpha,\beta))$

$\frac{d}{d\theta}\log(L(\theta))$

$= \frac{n}{\theta} + \frac{(x_1+x_2+\dots+x_n-n)}{1-\theta} * -1 + \frac{\alpha-1}{\theta} + \frac{\beta-1}{1-\theta} * -1 = 0$

$\frac{n+\alpha-1}{\theta} - \frac{x_1+\dots+x_n-n+\beta-1}{1-\theta} = 0$

$n+\alpha-1-n\theta-\alpha\theta+\theta - \sum x_i\theta + n\theta - \theta\beta + \theta = 0$

$n+\alpha-1 = \theta(\sum x_i + \alpha + \beta - 2)$

$$\theta = \frac{n+\alpha-1}{\sum x_i + \alpha + \beta - 2}$$

Date
Page

## Module-4

Clustering

Partition — K-means.

Hierarchical — Agglomerative

Distance metrics

Euclidian distance

$d(x,y) = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$  $(x_1, y_1) = (5,10)$

* Euclidian is a diagonal distance

Manhattan distance  City block distance

$d(x,y) = |x_2 - x_1| + |y_2 - y_1|$.

K Means Cluster

k → no. of clusters

mean of the similarity

by default - Euclidian distance

1) Select k random seeds.
2) Identify distance of all other pts with these seeds
3) Cluster the pts with least Euclidian dist
4) Take centroid of each cluster
5) Take centroid as the new seed
6) Continue iteration till cluster don't change

Q. Use K means clustering algorithm to divide following data into 2 clusters using Euclidean distances

| $x_1$ | 1 | 2 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $x_2$ | 1 | 1 | 3 | 2 | 3 | 5 |

| Point | Seed1 (2,1) | Seed2 (2,3) | Cluster |
|---|---|---|---|
| (1,1) | 1 | $\sqrt{2}$ | $C_1$ |
| (2,1) | 0 | 2 | $C_1$ |
| (2,3) | 2 | 0 | $C_2$ |
| (3,2) | $\sqrt{2}$ | $\sqrt{2}$ | $C_2$ |
| (4,3) | $\sqrt{8}$ | 2 | $C_2$ |
| (5,5) | 5 | $\sqrt{13}$ | $C_2$ |

New Centroid: (1.5, 1) (3.5, 3.25).

| Point | $C_1$ (1.5,1) | $C_2$ (3.5, 3.25) | Cluster |
|---|---|---|---|
| (1,1) | 0.5 | 3.36 | $C_1$ |
| (2,1) | 0.5 | 2.70 | $C_1$ |
| (2,3) | ~~1.12~~ 2.06 | 1.5 | $C_2$ |
| (3,2) | 1.8 | 1.3 | $C_2$ |
| (4,3) | ~~0.55~~ | 0.55 | $C_2$ |
| (5,5) | 5.3 | 2.30 | $C_2$ |

$C_1 = \{(1,1), (2,1)\}$

$C_2 = \{(2,3), (3,2), (4,3), (5,5)\}$

* In categorical dataset, k-means does not work

Q. Use k-means alg & Euclidean dist to cluster foll samples into 3 clusters

$A_1$ (2,10) $A_2$ (2,5) $A_3$ (8,4) $A_4$ (5,8) $A_5$ (7,5)
$A_6$ (6,4) $A_7$ (1,2) $A_8$ (4,9)

The distance matrix based on Euclidean dist is

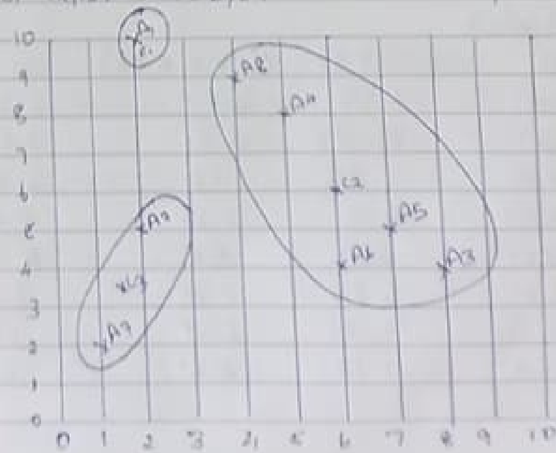
| | $A_1$ | $A_2$ | $A_3$ | $A_4$ | $A_5$ | $A_6$ | $A_7$ | $A_8$ |
|---|---|---|---|---|---|---|---|---|
| $A_1$ | 0 | $\sqrt{25}$ | $\sqrt{36}$ | $\sqrt{13}$ | $\sqrt{50}$ | $\sqrt{52}$ | $\sqrt{65}$ | $\sqrt{5}$ |
| $A_2$ | | 0 | $\sqrt{37}$ | $\sqrt{18}$ | $\sqrt{25}$ | $\sqrt{17}$ | $\sqrt{10}$ | $\sqrt{20}$ |
| $A_3$ | | | 0 | $\sqrt{25}$ | $\sqrt{2}$ | $\sqrt{2}$ | $\sqrt{53}$ | $\sqrt{41}$ |
| $A_4$ | | | | 0 | $\sqrt{13}$ | $\sqrt{17}$ | $\sqrt{52}$ | $\sqrt{2}$ |
| $A_5$ | | | | | 0 | $\sqrt{2}$ | $\sqrt{45}$ | $\sqrt{25}$ |
| $A_6$ | | | | | | 0 | $\sqrt{29}$ | $\sqrt{29}$ |
| $A_7$ | | | | | | | 0 | $\sqrt{58}$ |
| $A_8$ | | | | | | | | 0 |

Suppose that initial seeds are $A_1$, $A_4$ & $A_7$. Run k-means algorithm for 1 epoch only. At end of 1 epoch show
a) The new clusters
b) Centers of new clusters
c) Draw 10x10 space with all pts & show clusters after 1st epoch & new centroid
d) How many more iterations are need to converge? Draw result for each epoch

| Pt | $A_1$(2,10) | $A_4$(5,8) | $A_7$(1,2) | Cluster |
|---|---|---|---|---|
| (2,10) | 0 | $\sqrt{13}$ | $\sqrt{65}$ | $C_1$ |
| (2,5) | $\sqrt{25}$ | $\sqrt{18}$ | $\sqrt{10}$ | $C_3$ |
| (8,4) | 6 | 5 | $\sqrt{53}$ | $C_2$ |
| (5,8) | $\sqrt{13}$ | 0 | $\sqrt{52}$ | $C_2$ |
| (7,5) | $\sqrt{50}$ | $\sqrt{13}$ | $\sqrt{45}$ | $C_2$ |
| (6,4) | $\sqrt{52}$ | $\sqrt{17}$ | $\sqrt{29}$ | $C_2$ |
| (1,2) | $\sqrt{65}$ | $\sqrt{52}$ | 0 | $C_3$ |
| (4,9) | $\sqrt{5}$ | $\sqrt{2}$ | $\sqrt{58}$ | $C_2$ |

a) $C_1$ {$A_1$} $C_2$ {$A_3$, $A_4$, $A_5$, $A_6$, $A_8$} $C_3$: {$A_2$, $A_7$}

b) centers of new cluster
$C_1$ = (2,10) $C_2$ (6,6) $C_3$ (1.5, 3.5)

c)


Epoch 2

d) 

| Pt | C1(2,10) | C2(6,6) | C3(1.5,3.5) | Cluster |
|---|---|---|---|---|
| | ~~0~~ | ~~5.6~~ | ~~6.5~~ | ~~C1~~ |
| (2,10) | 0 | 5.6 | 6.5 | C1 |
| (2,5) | 5 | 4.1 | 1.5 | C3 |
| (8,4) | 8.4 | 2.8 | 6.5 | C2 |
| (5,8) | 3.6 | 2.23 | 5.7 | C2 |
| (7,5) | 7.07 | 1.414 | 5.7 | C2 |
| (6,4) | 5.65 | 2 | 4.5 | C2 |
| (1,2) | 8.06 | 6.4 | 1.58 | C3 |
| (4,9) | 2.23 | 3.6 | 6.04 | C1 |

C1 {A1 A8} C2 {A3 A4 A5 A6} C3 {A2 A7}

~~C1 (3,9.5)~~ ~~C2 (7,4.33)~~ ~~C3 (1.5~~

C1 (3, 9.5) C2 (6.5, 5.25) C3 (1.5, 3.5)

Epoch 3

| Pt | C1(3,9.5) | C2 (6.5, 5.25) | C3(1.5,3.5) | Cluster |
|---|---|---|---|---|
| (2,10) | 1.12 | 6.54 | 6.52 | C1 |
| (2,5) | ~~4.24~~ 4.61 | 4.51 | 1.58 | C3 |
| (8,4) | 7.43 | 1.95 | 6.52 | C2 |
| (5,8) | 2.5 | 3.13 | 5.70 | C1 |
| (7,5) | 6.02 | 0.56 | 5.70 | C2 |
| (6,4) | 6.27 | 1.35 | 4.53 | C2 |
| (1,2) | 7.76 | 6.39 | 1.58 | C3 |
| (4,9) | 1.12 | 4.61 | 6.04 | C1 |

C1 {A1, A4, A8} C2 {A3 A5 A6} C3 {A2, A7}

C1 (3.67, 9) C2 (7, 4.33) C3 (1.5, 3.5)

Epoch 4

| Pt | C1(3.67,9) | C2(7,4.33) | C3(1.5,3.5) | Cluster |
|---|---|---|---|---|
| (2,10) | 1.94 | 7.56 | 6.52 | C1 |
| (2,5) | 4.33 | 5.04 | 1.58 | C3 |
| (8,4) | 6.61 | 1.05 | 6.52 | C2 |
| (5,8) | 1.66 | 4.18 | 5.70 | C1 |
| (7,5) | 5.20 | 0.67 | 5.70 | C2 |
| (6,4) | 5.52 | 1.05 | 4.53 | C2 |
| (1,2) | 7.49 | 6.44 | 1.58 | C3 |
| (4,9) | 0.33 | 5.55 | 6.04 | C1 |

C1 {A1, A4, A8} C2 {A3, A5, A6} C3 {A2, A7}

Same as epoch 3 ∴ cluster centers are:

(3.67, 9) (7, 4.33) (1.5, 3.5)

2 more iteration needed

Hierarchical Clustering

* Bottom-Up Agglomerative Clustering

Start with each object in a sep cluster & repeat
- Joins the most similar pair of clusters
- Update the similarity of new cluster to other clusters until there is only one cluster
- Greedy - less accurate but simple, typically computationally expensive

Eg: AGNES

→ Top-Down divisive

Start with all data in a single cluster & repeat
- Split each cluster into 2 using a partition based algorithm until each object is a sep cluster

Eg: DIANA

Bottom-up Agglomerative Clustering

→ Single-link
Nearest Neighbour : similarity b/w their closest members

* Complete link
Farthest Neighbour : similarity b/w their farthest ~~m~~ members.

* Centroid - Similarity b/w center of gravity
* Average - Average similarity of all cross-cluster pair

Q. Use the dist matrix in Table 1 to perform single link & complete link hierarchical clustering

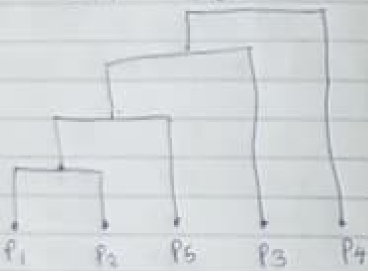
| | $P_1$ | $P_2$ | $P_3$ | $P_4$ | $P_5$ |
|---|---|---|---|---|---|
| $P_1$ | 0.00 | 0.10 | 0.41 | 0.55 | 0.35 |
| $P_2$ | 0.10 | 0.00 | 0.64 | 0.47 | 0.98 |
| $P_3$ | 0.41 | 0.64 | 0.00 | 0.44 | 0.85 |
| $P_4$ | 0.55 | 0.47 | 0.44 | 0.00 | 0.76 |
| $P_5$ | 0.35 | 0.98 | 0.85 | 0.76 | 0.00 |

Ans: Single-Linkage

| | $P_{12}$ | $P_3$ | $P_4$ | $P_5$ |
|---|---|---|---|---|
| $P_{12}$ | 0.00 | 0.41 | 0.55 | 0.35 |
| $P_3$ | 0.41 | 0.00 | 0.44 | 0.85 |
| $P_4$ | 0.55 | 0.44 | 0.00 | 0.76 |
| $P_5$ | 0.35 | 0.85 | 0.76 | 0.00 |

| | $P_{125}$ | $P_3$ | $P_4$ |
|---|---|---|---|
| $P_{125}$ | 0.00 | 0.41 | 0.55 |
| $P_3$ | 0.41 | 0.00 | 0.44 |
| $P_4$ | 0.55 | 0.44 | 0.00 |

| | $P_{1235}$ | $P_4$ |
|---|---|---|
| $P_{1235}$ | 0.00 | 0.44 |
| $P_4$ | 0.44 | 0.00 |




Algorithm

1. Compute the proximity matrix, if necessary
2. Repeat
3. Merge the closest two clusters
4. Update the proximity matrix to reflect the proximity b/w new cluster & original clusters
5. until. Only one cluster remains

Complete linkage

| | $P_1 P_2$ | $P_3$ | $P_4$ | $P_5$ |
|---|---|---|---|---|
| $P_1 P_2$ | 0 | 0.64 | 0.55 | 0.98 |
| $P_3$ | 0.64 | 0 | 0.44 | 0.85 |
| $P_4$ | 0.55 | 0.44 | 0 | 0.76. |
| $P_5$ | 0.98 | 0.85 | 0.76 | 0. |

| | $P_{125}$ | $P_3$ | $P_4$ |
|---|---|---|---|
| $P_{125}$ | 0 | 0.85 | 0.76 |
| $P_3$ | 0.85 | 0 | 0.44 |
| $P_4$ | 0.76. | 0.44 | 0. |

| | $P_{1235}$ | $P_4$ |
|---|---|---|
| $P_{1235}$ | 0 | 0.76. |
| $P_4$ | 0.76. | 0. |